

LYDIA PEDERSEN

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PROFESSIONAL SUMMARY

Documentation engineer with 20 years of experience across physics research, hardware validation, and AI-powered content systems. I design and build agentic pipelines that automate documentation creation, quality assurance, link governance, and analytics-driven prioritization. Every system uses controlled, quasi-deterministic workflows with rigorous testing, structured auditing, and quantified output quality. Every result is evidence-backed: outputs are correlated, thresholds are calibrated, and regressions are caught before they ship. My career reflects a continuous progression from validating silicon to validating content, applying the same engineering discipline to documentation that hardware teams apply to compliance testing.

PROFESSIONAL EXPERIENCE

Senior Technical Writer

Gravitee · Remote

Jun 2023 – Jul 2026

Designed and built a suite of AI-powered documentation engineering pipelines from the ground up, transforming manual documentation workflows into automated, auditable, and reproducible systems. All pipelines are developed through AI-assisted orchestration (Claude, Gemini), open-sourced on GitHub, fully tested with pytest, and produce structured audit trails with quantified quality metrics.

AGENTIC PIPELINE DEVELOPMENT

- **AI Doc Writer Agent:** Multi-stage content generation pipeline that converted PR diffs and Jira tickets into publication-ready documentation. Featured fidelity scoring, Best-of-N evaluation with adaptive scaling, a 14-function guard cascade to prevent hallucination and editorial fabrication, 20-step post-processing, and multi-stage recovery. Output quality was measured through containment-based deduplication metrics and cross-file consistency checks.
- **Analytics Deflection Engine:** Abstracted a 14,182-line monolith into a modular, product-agnostic 9-step pipeline that transformed raw AI assistant analytics into ranked, privacy-safe documentation improvement reports. Featured a 200-combination parameter sweep for self-correction, 3-signal priority scoring (impact, pain, breadth), a 21-step text cleaning cascade, k-anonymity privacy model, and automated Jira task generation. Surfaced what users actually struggled with so documentation teams could close knowledge gaps systematically.
- **Link Governance Engine:** 19-category link classifier with a safety-gated autofix system, page-version index validation, and external link health checking. Automated broken link detection, classification, and remediation across versioned documentation sites with deterministic, auditable fix decisions.
- **Semantic Redirect Engine:** 4-stage redirect pipeline (Extract, Embed, Match, Emit) using embedding-based semantic similarity with a penalty stack, token intelligence system, and multi-tier fallback cascade. Generated intelligent cross-version page mappings that preserved user navigation across documentation releases.

AI DEVELOPMENT METHODOLOGY

- Built all pipelines through AI-assisted development using agentic and prompt engineering patterns. Used model selection strategically to optimize for token budgets and rate limits. Maintained fresh context through checkpoints, structured handoffs, and living documents that preserved continuity across conversations. All work was version-controlled and followed a reproducible, non-destructive development pattern.

DOCUMENTATION ENGINEERING & STRATEGY

- Automated headless browser-based screenshot and UI capture, eliminating manual image maintenance. Integrated PR diff analysis for change-aware content generation synchronized with product releases.
- Elevated documentation to a first-class engineering deliverable. Coauthored team processes, templates, strategies, and roadmaps. Redesigned documentation site information architecture through goal-centric user journey analysis. Integrated documentation into the internal business review cycle via Jira and Slack.

Technical Writer

Entando · Remote

Owned developer documentation for a Kubernetes-based application composition platform. Verified all content in cloud and Multipass VM environments with hands-on testing of MFE/MS generation, product UIs, and CLI tooling.

- Maintained a VuePress static documentation site with Markdown/Vue styling. Created and owned the company glossary and style guide, standardizing terminology across all customer-facing materials.
- Authored a high-impact white paper used to secure new investors. Performed content and language review of marketing, developer advocacy, and partner communications. Coached non-writers in documentation best practices.

Oct 2021 – Mar 2023

Hardware Validation Engineer

Amazon Lab126 · Sunnyvale, CA

Led high-speed interface compliance testing (USB, SATA, HDMI) from proof-of-concept through production certification for consumer devices. Owned acceptance criteria, test plans, data collection, and design feedback. This is the same systematic validation methodology I now apply to documentation pipeline output.

- Developed test automation scripts and custom tooling for data collection, processing, and analysis across evolving product lines. Built repeatable, instrumentation-aware test methodologies for comprehensive coverage.
- Managed vendor relationships for test equipment procurement and support. Maintained industry engagement to track standards revisions and emerging methodologies.

Apr 2017 – Sep 2021

Senior Hardware Engineer

Oracle · Santa Clara, CA

TX/RX high-speed interface characterization: jitter analysis, signal quality measurements, and PLL tuning. Developed test scripts to automate data collection, DUT parameter control, and result formatting. Applied jitter tolerance methodologies to signal integrity analysis and design feedback. Benchtop optimization including calibration, S-parameters, and post-processing.

Jul 2014 – Mar 2017

Senior Hardware Engineer

LSI Logic (now Broadcom) · Milpitas, CA

Compliance testing across PCIe, SATA, SAS, Fibre Channel, and IEEE 802.3 standards at speeds up to 32 Gbps. Early adopter of PAM4 benchtop solutions. Contributed data collection and methodology experimentation to the IEEE 802.3bj standards committee. Developed test scripts and equipment drivers to automate measurements, stress-test functionality, and perform statistical analysis.

Apr 2008 – Jun 2014

EARLIER CAREER

Consultant, LipidX Technologies (2007–2008): Statistical modeling, data mining, automated neural net and regression model validation.

Product Applications Engineer, LSI Logic (2006–2007): Built data analysis automation tools (VBA). Serial communications compliance testing (PCIe, SAS, Fibre Channel).

Research Assistant, UC Irvine (2005–2006): Senior thesis on optical modeling of sub-Antarctic neutrino detection. Software simulation and experimental correlation for an international physics campaign.

EDUCATION

B.S. Physics, University of California, Irvine

Graduated with Honors (3.7 GPA) · UC Regents Scholar · National Physics Honors Society · Campuswide Honors Program

2002 – 2006

TECHNICAL SKILLS

AI & ML	Agentic pipeline architecture, LLM orchestration (Claude, Gemini), prompt engineering (agentic and structured patterns), AI-assisted development, transformer architecture (embeddings, positional encoding, Q/K/V attention, activation layers, backpropagation), RAG, semantic search, fidelity scoring, Large Language Models (LLMs), Model Context Protocol (MCP), Agentic Orchestration
LANGUAGES & TOOLS	Python, YAML, JSON, Markdown, Bash/Zsh, Git/GitHub, VS Code, Jira, pytest
DOCUMENTATION	Docs-as-code, static site generators (GitBook, VuePress), API documentation, information architecture, content strategy, style guides, versioned documentation, glossary management
DATA & ANALYSIS	Statistical modeling, parameter sweep optimization, regression analysis, output correlation, threshold calibration, data pipeline design, privacy modeling (k-anonymity)
INFRASTRUCTURE	Docker, Kubernetes, CI/CD workflows, cloud environments, Linux CLI, headless browser automation